

HW5: Model Comparison for House Prices

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- [real_estate_prices.csv](#)

The response is Price, the sales price of a house in thousands of dollars. Size is home size in hundreds of square feet. Rating is an overall niceness rating from 1 to 10.

Fit and compare these models:

Model 1: $Price = \beta_0 + \beta_1 Size + \beta_2 Rating + \epsilon$.

Model 2: $Price = \beta_0 + \beta_1 Size + \beta_2 Rating + \beta_3 Rating^2 + \epsilon$.

Problem 1(a): Scatter Plots

Create scatter plots of Price versus Size and Price versus Rating. Discuss why these plots make the two models reasonable candidates.

Problem 1(b): Fit Both Models and Interpret Coefficients

Fit Model 1 and Model 2. Interpret the regression coefficients in context. For Model 2, remember that the effect of Rating is not constant because the model contains $Rating^2$.

Problem 1(c): Individual Coefficient Tests

For each model, test the significance of each slope coefficient at level 0.05. State hypotheses, p-values or test statistics, and conclusions.

Problem 1(d): Overall Regression Test

For each model, test whether the overall regression model is significant at level 0.05.

Problem 1(e): Prediction at Size = 2000 Square Feet and Rating = 8

For an individual house with size 2000 square feet and rating 8, compute the point estimate, 95% confidence interval for the mean response, and 95% prediction interval for an individual house under both models.

Remember: Size is measured in hundreds of square feet, so 2000 square feet corresponds to $Size = 20$.

Problem 1(f): Compare Using Adjusted R-Squared

Compare the two models using adjusted R^2 . Which model would you prefer by this criterion?

Problem 1(g): Compare Using CI and PI

Compare the confidence intervals and prediction intervals from the two models. Which model gives narrower intervals at the requested predictor setting? Does this agree with your adjusted R^2 comparison?

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.